



Course : R-VIJETA (R-JP)

PHYSICS DPP

DAILY PRACTICE PROBLEM

REVISION DPP NO. # 06

Target : JEE(Advanced) 2023

DPP Syllabus :

WAVES, WAVE OPTICS & MODERN PHYSICS

DPP No. : 06

Maximum Marks : 170

Total Time : 160 Min.

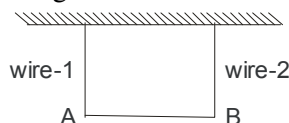
One or more than one may be correct option ('-1' Negative marking) Q.1 to 20; (3 Marks; 2 Min.) [60, 40]

Subjective questions ('-1' Negative marking) Q.21 to Q.40; (4 Marks; 5 Min.) [80, 100]

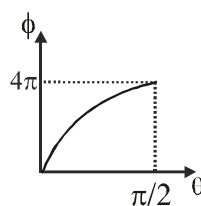
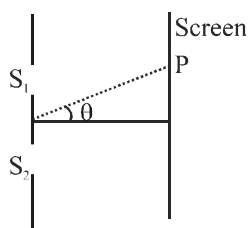
Single Choice Objective ('-1' Negative marking) Q.41 to Q.46 (3 Marks; 2 Min.) [18, 12]

Comprehension ('-1' Negative Marking) Q.47 to Q.50 (3 marks 2 min.) [12, 08]

1. A rod of length 0.3 m having variable linear mass density from A to B as $\lambda = \lambda_0 x$ (x is distance from A in meter), where $\lambda_0 = 100 \text{ kg/m}^2$ is suspended by two light wires of same length. Ratio of their linear mass density is 2 : 9. Then which of the following is/are correct :



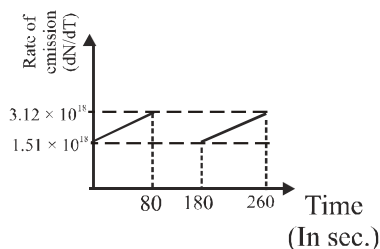
- (A) Ratio of wave speed in wire-1 to wire-2 is 3 : 2.
 (B) Ratio of wave speed in wire-1 to wire-2 is 3 : 1.
 (C) Second harmonic in wire-1 has same frequency as third harmonic in wire-2.
 (D) Third overtone in wire-1 has same frequency as fifth overtone in wire-2.
2. In Young's double slits experiment set up, the graph of the phase difference (ϕ) between both waves at a point P on the screen versus the angular position θ of point P on the screen is as shown in the figure. Then,



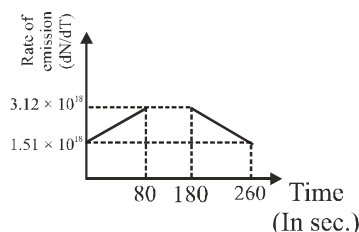
- (A) The distance $S_1 S_2 = 2\lambda$.
 (B) There are total 4 minima on the screen.
 (C) The first maxima above the centre is at $\theta = \frac{\pi}{6}$.
 (D) At $\theta = \sin^{-1}\left(\frac{3}{4}\right)$, intensity is minimum.

3. A light of wavelength λ is incident on a metal plate of work function $\phi = 2\text{ eV}$. The wavelength λ varies with time as $\lambda = 3000 + 40 kt$, where $\lambda(\text{\AA})$, t (second) and $k = 1 \text{ \AA}/\text{sec}$. The power incident on metal sheet is 100 W . This signal is switched on and off for time interval of 2 minute and 1 minute respectively. Each time the signal is switched on the λ starts from initial value of $3000 \text{ \AA}$. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1% . ($hc = 12400 \text{ eV\AA}$) -

(A) The variation of rate of emission of photoelectron $\left(\frac{dN}{dt}\right)$ versus time is.



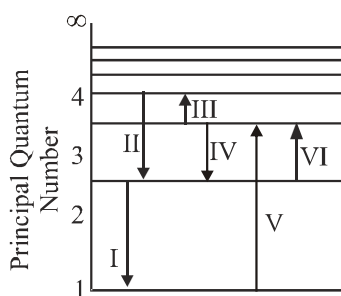
(B) The variation of rate of emission of photoelectron $\left(\frac{dN}{dt}\right)$ versus time is.



(C) The total number of photo electrons ejected in one hour is 3.71×10^{21} .

(D) The total number of photo electrons ejected in one hour is 3.71×10^{23} .

4. The figure above shows an energy level diagram for the hydrogen atom. Several transitions are marked as I, II, III, _____ . The diagram is only indicative and not to scale.



- (A) The transition in which a Balmer series photon absorbed is VI.
 (B) The wavelength of the radiation involved in transition II is approximately 486 nm .
 (C) IV transition will occur when a hydrogen atom is irradiated with radiation of wavelength 103 nm .
 (D) IV transition will emit the longest wavelength line in the visible portion of the hydrogen spectrum.

5. Radioactive nuclei A simultaneously decays in P and Q with decay constant 2λ and λ respectively. Another radioactive nucleus B decays in Q with decay constant 3λ . Initially number of active nuclei of A and that of B are equal (N_0). Select correct relation for any later moment t .

(A) $N_A = N_B = N_0 e^{-3\lambda t}$

(B) $N_P = N_Q = \frac{N_0}{3} (1 - e^{-3\lambda t})$

(C) $N_P = 2N_Q = \frac{N_0}{3} (1 - e^{-3\lambda t})$

(D) $2N_P = N_Q = \frac{4N_0}{3} (1 - e^{-3\lambda t})$

6. If one of the slit of a standard YDSE apparatus is covered by a thin glass slab so that it allows only one half of the light intensity of the other then :

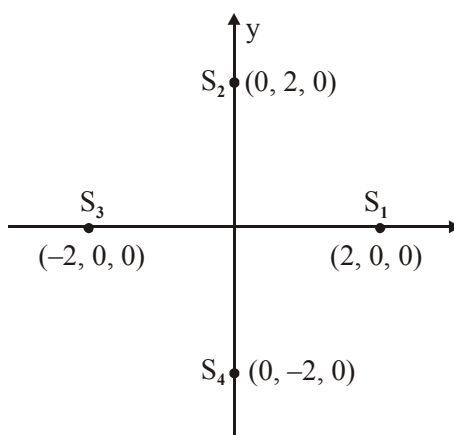
(A) Resultant intensity at centre of screen will decrease.

(B) Resultant intensity at dark fringe will increase.

(C) The fringe width will remain unchanged.

(D) The fringe pattern will shift towards the covered slit.

7. Four point sources each of frequency (f) are fixed at their position as shown in the figure. An observer moving with constant velocity $\vec{V}$ is crossing origin. 'N' is number of different frequencies observed by the observer at the moment when it is crossing origin.



(A) If $\vec{v} = 5 \hat{i}$ then $N = 3$

(B) If $\vec{v} = 5(\hat{i} + \hat{j})$ then $N = 4$

(C) If $\vec{v} = 5 \hat{k}$ then $N = 1$

(D) If $\vec{v} = (\hat{i} + \hat{j} + \hat{k})$ then $N = 2$

8. ${}_{92}\text{U}^{235}$ is a α -active nucleus. For large quantity of sample size, select the correct option (s) :

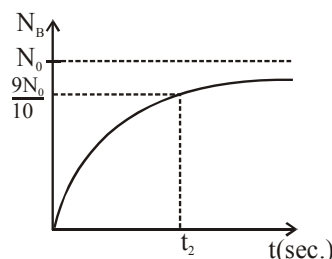
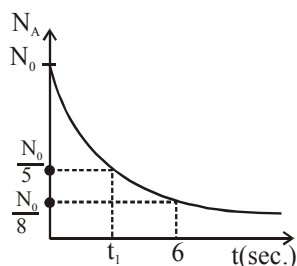
(A) The probability of a nucleus disintegration during one second interval remains constant for all time.

(B) Quite an appreciable quantity will remain even after the average life time.

(C) The probability of a nucleus disintegration during one second is lower in first half-life and greater in second half-life.

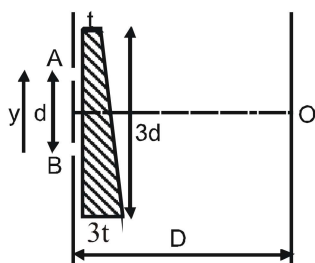
(D) The energy of the emitted α -particles is less than the disintegration energy of U^{235} .

9. A steel wire is rigidly fixed at both ends. Its length, mass and cross sectional area are 1 m, 0.1 kg and 10^{-6} m^2 respectively. Then the temperature of the wire is lowered by 20°C . If the transverse waves are setup by plucking the wire at 0.25 m from one end and assuming that wire vibrates with minimum number of loops possible for such a case.
[Coefficient of linear expansion of steel = $1.21 \times 10^{-5}/^\circ\text{C}$ and Young's modulus = $2 \times 10^{11} \text{ N/m}^2$]
(A) The tension in the wire will be 48.4 N. (B) The number of loops will be 2.
(C) The frequency of vibration will be 22 Hz. (D) The frequency of vibration will be 44 Hz.
10. An X-ray tube is operating at 50 kV and 20 mA. The target material of the tube has mass of 1 kg and specific heat $495 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$. One percent of applied electric power is converted into X-rays and the remaining energy goes into heating the target. Then :
(A) a suitable target material must have high melting temperature.
(B) a suitable target material must have low thermal conductivity.
(C) The average rate of rise of temperature of the target would be 2°C/sec .
(D) The minimum wavelength of X-rays emitted is about $0.25 \times 10^{-10} \text{ m}$.
11. The electron in hydrogen atom makes a transition $n_1 \rightarrow n_2$ where n_1 and n_2 are the principal quantum number of two states. Assuming the Bohr model to be valid, the time period of the electron in the initial state is eight times that in the final state. The possible value of n_1 and n_2 are :
(A) $n_1 = 2$ and $n_2 = 1$ (B) $n_1 = 8$ and $n_2 = 2$ (C) $n_1 = 8$ and $n_2 = 1$ (D) $n_1 = 6$ and $n_2 = 3$
12. In a decay process A decays to B, $A \rightarrow B$. Two graphs of number of nuclei of A and B versus time is given then :



- (A) $t_2 - t_1 = 4$ (B) $t_2 - t_1 = 2$ (C) $t_1 = 2 \log_2 5$ (D) $t_2 = \log_2 100$
13. A car moves towards a hill with speed v_c . It blows a horn of frequency f which is heard by an observer following the car with speed v_0 . The speed of sound in air is v .
(A) the wavelength of sound reaching the hill is $\frac{v}{f}$.
(B) the wavelength of sound reaching the hill is $\frac{v - v_c}{f}$.
(C) The wavelength of sound of horn directly reaching the observer is $\frac{v + v_c}{f}$.
(D) the beat frequency observed by the observer is $\frac{2v_c(v + v_0)f}{v^2 - v_c^2}$.

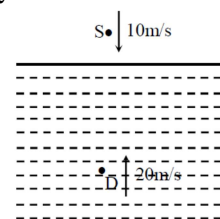
14. A neutron having kinetic energy E collides head on with a stationary hydrogen atom in ground state, which is initially at rest but free to move. Assume the mass of the neutron to be equal to that of H atom. Use the energy of the n^{th} state for H atom as $E_1 = -13.6 \text{ eV}$; $E_2 = -3.4 \text{ eV}$; $E_3 = -1.5 \text{ eV}$. (whether collision is elastic or inelastic, depends on the energy of the neutron, and in case of collision they can be treated as ordinary particles) now choose the correct options :
- (A) The collision will be elastic if E is less than 20.4 eV .
 (B) If $E = 24.2 \text{ eV}$, the H atom may get excited to $n = 4$.
 (C) If $E < 20.4 \text{ eV}$ the kinetic energy of the neutron will become zero after the collision.
 (D) If $E > 20.4 \text{ eV}$ the kinetic energy of the neutron will become zero after the collision.
15. In a standard YDSE two slits are covered by a wedge shape film of refractive index μ as shown in the figure. The thickness of film varies linearly from t to $3t$ from one end as shown in the figure. The distance between the two slits is d . The screen is at distance D from slits. White light is used in experiment. Choose the correct option (s), ($d \ll D$). Select correct alternative(s) :



- (A) Central maxima is formed at distance $\frac{2t(\mu-1)D}{3d}$ below O.
 (B) If the film is displaced $\frac{d}{2}$ along $+y$ axis the central maxima will shift away from O.
 (C) If the film is displaced $\frac{d}{2}$ along $-y$ -axis the central maxima will shift away from O.
 (D) If the film is displaced $\frac{d}{2}$ along $+y$ axis the fringe width will remain same.
16. The correct statements among the following are :
 (Consider only normal incidence) (P = Power, I = Intensity, E = energy, c = speed of light)
- (A) Pressure exerted by photons for perfectly reflecting surface is $\frac{2I}{c}$.
 (B) Force exerted by photons on a perfectly reflecting surface is $\frac{2P}{c}$.
 (C) Impulse applied by photon on a perfectly reflecting surface is $\frac{2E}{c}$.
 (D) Force exerted by photons on a perfectly reflecting surface is $\frac{P}{c}$.

17. A person riding on a source 'S' of sound (frequency 48 Hz) moves in a medium (in which velocity of sound is 310 m/s) with speed 10m/s as shown towards a liquid reservoir (in which velocity of sound is 1240 m/s). A detector 'D' in the reservoir moves towards the free surface of the reservoir (along line SD) at a speed 20m/s. The sound produced by 'S' gets reflected and transmitted at the free surface. Mark the incorrect statement(s) among the following. Assuming the liquid in the reservoir stays still and calm.

- (A) Wavelength of sound received by detector D is 25 m
 (B) Beat frequency registered by the person is 1.6 Hz
 (C) Frequency of sound detected by detector D is 56 Hz
 (D) Frequency of sound detected by detector D is 52.4 Hz



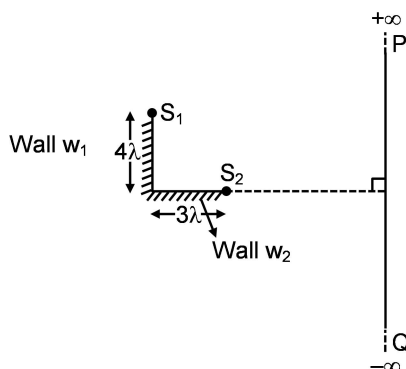
18. An organ pipe of length 80 cm is open at $x = 0$ and closed at $x = 80$ cm. Speed of sound in the air column is 320 m/sec. If standing waves are generated in the closed organ pipe, then the correct equation(s) of standing waves is/are (Here s = longitudinal displacement, A = displacement amplitude, P_{ex} = pressure excess, P_0 = pressure excess amplitude, t = time) (Neglect the end correction.)

- (A) $s = A \cos\left(\frac{5\pi}{4}x\right) \sin(400\pi t)$ (B) $s = A \cos\left(\frac{15\pi}{8}x\right) \cos(600\pi t)$
 (C) $P_{ex} = P_0 \cos\left(\frac{5\pi}{8}x\right) \sin(200\pi t)$ (D) $P_{ex} = P_0 \sin\left(\frac{25\pi}{8}x\right) \cos(1000\pi t)$

19. At $t = 0$, a sample of radionuclide A has the same decay rate as a sample of radionuclide B has at $t = 60$ min. The disintegration constants of A and B are λ_A and λ_B respectively, with $\lambda_A < \lambda_B$. Choose the correct option(s) :

- (A) The half life of radionuclide A is greater than that of B.
 (B) At $t = 60$ min, number of atoms in sample of material A is greater than that of sample B.
 (C) The two samples will never have the same decay rate simultaneously.
 (D) After certain time, the two samples will have the same decay rate simultaneously for an instant.

20. Consider 2 identical coherent sources S_1 & S_2 kept as shown in figure. Screen is very large. Wavelength of light emitted by source is λ .

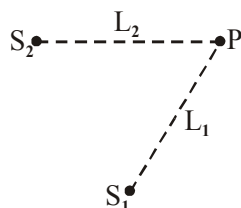


Assume no reflection and transmission from wall w_1 and w_2 . Choose correct option(s) regarding the interference pattern observed on straight line PQ.

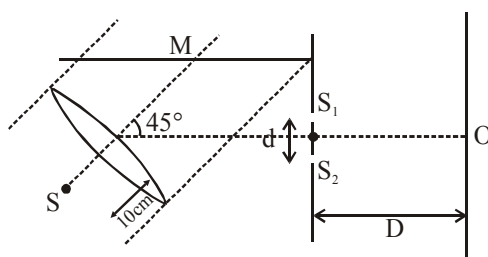
- (A) Total number of maxima observed on line PQ above central maxima is 3.
 (B) Total number of maxima observed on line PQ below central maxima is 3.
 (C) Total number of maxima observed on line PQ is 7.
 (D) Total number of maxima observed on line PQ is 9.

21. The present day abundances of the isotopes U^{238} and U^{235} are in the ratio of 128 : 1. They have half lives of 4.5×10^9 years and 7×10^8 years respectively. If age of earth is $\frac{7x}{76} \times 10^9$ years, then calculate x : (Assume equal amount of each isotope existed at the time of formation of the earth).
22. In YDSE with monochromatic light, fringes are obtained on the screen placed at some distance from the slits. If screen is moved by 5×10^{-2} m towards the slits, fringe width changes by 3×10^{-5} m. If separation between the slits is 10^{-3} m, if wavelength of light used is $100\text{y \AA}$, then y is : (Assume distance between the two slits is very small than the distance between the slits and screen)
23. A fork and a string of length 100 cm give 4 beats/s. The string is made shorter, without any change of tension, until it is in unison with the fork. Its new length is 99 cm. The frequency of the fork is 80P Hz. Find the value of P.
24. When the voltage applied to an X-ray tube increases from 10 kV to 20 kV the wavelength difference between the K_α line and the cut-off wavelength of the continuous X-ray spectrum increases by a factor of 3.0. Find atomic number of target metal. (Given that $\frac{1}{\text{Rydberg constant}} = 900\text{\AA}$).
25. In a hypothetical atom, negatively charged particle having a charge of magnitude 3 e and mass 3 m revolves around a proton. Here, e is the magnitude of electronic charge and m is the electronic mass. Mass of proton may be assumed to be much larger than that of the negatively charged particle, thus the proton is at rest. This "atom" obeys Bohr's postulate of quantization of angular momentum, that is $mvr = n\left(\frac{h}{2\pi}\right)$. It is given that for the first Bohr orbit of hydrogen atom total energy is $-E_0$. The momentum of an emitted photon when it makes a transition from the second excited state to ground state is $\frac{NE_0}{c}$, where 'c' is speed of light. Find the value of N.
26. If the frequency of K_α X-ray emitted from the element with atomic number 31 is f, if the frequency of K_α x-ray emitted from the element with atomic number 51, is given by $\frac{25}{x}f$, then x is (assume that screening constant for K_α is 1).
27. Subsonic aircraft flying in circular path of radius R in vertical plane with constant speed V. Speed of sound in medium is V_w . (assume the medium to be at rest.) Detector which can detect position of plane as well as frequency of sound emitted by plane lie below the centre of circular path at distance 2R from the centre. Whenever detector detects maximum frequency, its also detects that plane is nearest to detector. In this condition $\frac{V}{V_w} = \frac{11\alpha}{(11\alpha - 1)\sqrt{3}}$ Here α is an integer. Find α . ($\pi = 22/7$).

28. A hydrogen atom is in its excited state with energy -1.51 eV. The angular momentum of the electron in this state is $\frac{xh}{\pi}$ then the value of 'x' is :
29. An element X decays, first by positron emission and then two α -particles are emitted in successive radioactive decay. If the product nuclei has a mass number 229 and atomic number 89, the atomic number of element X is :
30. Two sound sources shown in the figure vibrate in phase. By moving S_1 along PS_1 consecutive minima are heard when $L_1 - L_2$ has values, 20 cm, 60 cm and 100 cm. If the frequency of sound source is n Hz. Then find the value of n . [Speed of sound is 325 m/s] :



31. Sinusoidal waves 5.00 cm in amplitude are to be transmitted along a string having a linear mass density equal to 4.00×10^{-2} kg/m. If the source can deliver a average power of 180 W and the string is under a tension of 100 N. The highest frequency at which the source can operate is f . Find f (take $\pi^2 = 10$) :
32. In Young's double slit experiment, distance between the slits is d and that between the slits and screen is D . Angle between principle axis of lens and perpendicular bisector of S_1 and S_2 is 45° . The point source S is placed at the focus of lens and aperture of lens is much larger than d . Assuming only the reflected light from plane mirror M is incident on slits, distance of central maxima from O will be X then find out $\frac{X}{D}$:



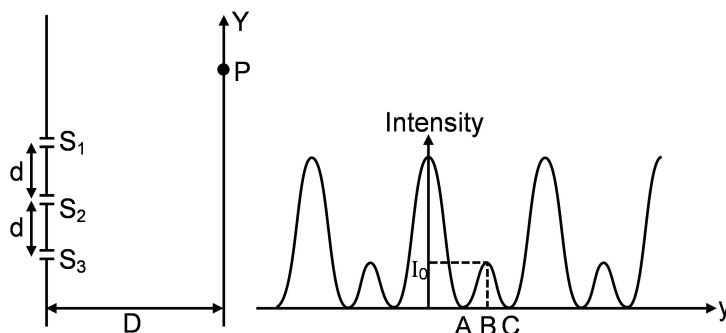
33. Two radioactive materials A and B have decay constants 5λ and λ respectively. Initially both A and B have the same number of nuclei. The ratio of the number of nuclei of A to that of B will be $\frac{1}{e}$, after the time $\frac{x}{8\lambda}$, then x is :

34. A load of 20 kg is suspended by a steel wire as shown in figure. Velocity of longitudinal waves is 20 times the velocity of the transverse waves in the same string when it is plucked. If area of cross section of the wire is $x \times 10^{-7} \text{ m}^2$. Find x (Y for steel is $19.6 \times 10^{10} \text{ N/m}^2$ and $g = 9.8 \text{ m/s}^2$) :



35. A metal surface has work function 1.1 eV. Photon of wavelength $6200 \text{ \AA}$ is absorbed by an electron which makes a collision with lattice before coming to surface. If it is observed that it comes out with kinetic energy 0.4 eV, then percentage loss of kinetic energy during collision with lattice is $x \%$ (approximately). Then find $\frac{x}{5}$:

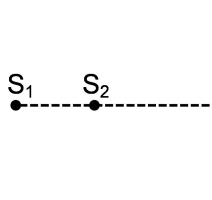
36. Consider the case of three sources in phase, similar to YDSE, as shown in the figure. At a distance $D \gg d$ from sources, the rays from the sources to a point P on screen are nearly parallel. A plot of intensity pattern on the screen far from the sources is also shown in the figure. If the distance between A and C is $\frac{2\lambda D}{Nd}$ then find the value of N.



37. Two forks A and B when sounded together produce 4 beats/sec. The fork A is in unison with 30 cm length of a sonometer wire and B is in unison with 25 cm length of the same sonometer wire at the same tension in same mode. If their frequency are f_A and f_B and $f_A + f_B = 11 \text{ N}$. Then find the value of N.
38. A car while travelling produces noise of intensity level of 94 decibel. At a particular point on road maximum permissible level of noise is 100 decibel. Find how many such identical cars can be allowed to pass through that point simultaneously if sound waves emitted by all cars are in same phase at that point? (Use $2 = 0.3$)

39. Tension in a uniform string of length L varies linearly with length. At one end A tension is T_0 and at other end B it is $16 T_0$. If mass per unit length of string is μ then time taken by a pulse to move from end A to end B is $\frac{\alpha}{5} L \sqrt{\frac{\mu}{T_0}}$. Here α is an integer. Find α .

40. Two coherent monochromatic point sources S_1 and S_2 are placed in front of an infinite screen as shown in figure. Wavelength of the light emitted by both the sources is λ . Initial phase difference between the sources is zero. Initially $S_1 S_2 = 2.5\lambda$ and the number of bright circular rings on the screen is n_1 . If the distance $S_1 S_2$ is increased and made 5.7λ , the number of bright circular rings becomes n_2 . The difference $(n_2 - n_1)$ is :



41. If the surface of a metal is exposed, by two different radiations, of wavelength $\lambda_1 = 350 \text{ nm}$ and $\lambda_2 = 450 \text{ nm}$. Maximum velocity of photoelectrons are observed to be in the ratio of $2 : 1$. Work function of the metal is :

(A) $2.84 \times 10^{-18} \text{ J}$ (B) $1.6 \times 10^{19} \text{ J}$ (C) $4 \times 10^{-19} \text{ J}$ (D) $2.4 \times 10^{-17} \text{ J}$

42. Half lives of two isotopes x and y of a material are known to be 2×10^9 years and 4×10^9 years respectively. A planet was formed with equal number of atoms of these isotopes. If planet currently contains 20% of x and 80% of Y by number then the current age of the planet is :

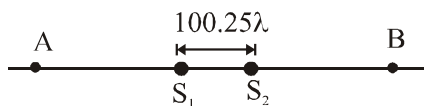
(A) 2×10^9 year (B) 4×10^9 year (C) 6×10^9 year (D) 8×10^9 year

43. An electron in hydrogen atom makes transition from one orbit to another. Magnitude of change in orbital angular momentum of the electron may be :

(A) h (B) $\frac{h}{\pi}$ (C) $\frac{h}{8\pi}$ (D) $\frac{h}{4\pi}$

44. S_1 and S_2 are two coherent sources of radiations separated by distance 100.25λ , where λ is the wave length of radiation. S_1 leads S_2 in phase by $\pi/2$. A and B are two points on the line joining S_1 and S_2 as shown in figure. The ratio of amplitudes of component waves from source S_1 and S_2 at A as well as at B are in ratio

$1 : 2$. Then the ratio of resulting intensity at A to that at B, $\left(\frac{I_A}{I_B}\right)$ is:



(A) ∞ (B) $1/9$ (C) 0 (D) 9

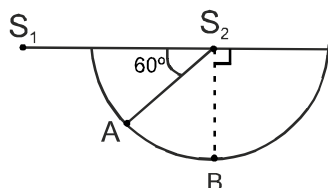
45. Light of wavelength 400 nm is incident continuously on a Cesium ball (work function 1.9 eV). The maximum potential to which the ball will be charged is :

(A) 3.1 V (B) 1.2 V (C) zero (D) infinite

46. To reduce the light reflected by the glass surface of a camera lens, the surface is coated with a thin layer of another material which has an refractive index ($\mu = 7/4$) smaller than that of glass. The least thickness of the layer, to ensure that light falling perpendicularly on the surface and having wavelengths, $\lambda_1 = 700\text{nm}$ and $\lambda_2 = 420\text{nm}$ will be weakly reflected for both wavelengths :
- (A) 200 nm (B) 300 nm (C) 400 nm (D) 500 nm

Paragraph for Questions no. 47 & 48

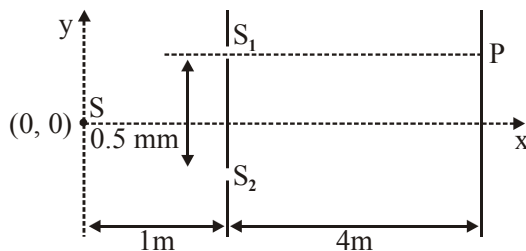
Figure shows two long line sources of sound S_1 and S_2 separated by a distance 4 m. The two sources are in same phase at all times. The sources emit same power and their lengths are also same. A detector moves along a circle with center at S_2 and radius 3m. The wavelength of the sound is 1 m. When it is at A the intensity of sound due to source S_2 only is I_0 .



47. The intensity of sound at A due to S_1 only is :
- (A) $\frac{3I_0}{\sqrt{13}}$ (B) $\frac{\sqrt{13}I_0}{3}$ (C) $\frac{9I_0}{13}$ (D) I_0
48. The intensity of sound at B due to S_1 and S_2 is :
- (A) $\frac{70}{25}I_0$ (B) $I_0 \left[\frac{8}{5} + 2\sqrt{\frac{3}{5}} \right]$ (C) $\frac{8I_0}{5}$ (D) $I_0 \left[\frac{8}{5} + 2\sqrt{\frac{9}{25}} \right]$

Paragraph for Question Nos. 49 to 50

Figure shows a Young's double slit experiment set-up. The source S of wavelength 4000Å oscillates along y-axis according to the equation $y = \sin \pi t$, where y is in millimeters and t is in seconds. The distance between two slits S_1 and S_2 is 0.5 mm.



49. The instant at which maximum intensity occurs at P for first time :
- (A) $\frac{1}{\pi} \sin^{-1} \left(\frac{59}{160} \right)$ (B) $\frac{1}{\pi} \sin^{-1} \left(\frac{27}{80} \right)$ (C) $\frac{1}{\pi} \sin^{-1} \left(\frac{59}{80} \right)$ (D) $\frac{1}{\pi} \sin^{-1} \left(\frac{27}{160} \right)$
50. The instant at which minimum intensity occurs at P for the first time :
- (A) $\frac{1}{\pi} \sin^{-1} \left(\frac{59}{160} \right)$ (B) $\frac{1}{\pi} \sin^{-1} \left(\frac{27}{80} \right)$ (C) $\frac{1}{\pi} \sin^{-1} \left(\frac{59}{80} \right)$ (D) $\frac{1}{\pi} \sin^{-1} \left(\frac{27}{160} \right)$

ANSWER KEY TO DPP # 06 (JEE-ADVANCED)

1.	(A,C,D)	2.	(A,B,C,D)	3.	(A,C)	4.	(A,B,D)	5.	(A,D)
6.	(A,B,C,D)	7.	(A,C,D)	8.	(A,B,D)	9.	(A,B,C)	10.	(A,C,D)
11.	(A,D)	12.	(B,C,D)	13.	(B,C,D)	14.	(A,C)	15.	(A,D)
16.	(A,B,C)	17.	(B,C,D)	18.	(B,D)	19.	(A,D)	20.	(A,D)
21.	63	22.	60	23.	5	24.	29	25.	24
26.	9	27.	2	28.	1.50	29.	94	30.	812.50
31.	42.43	32.	1	33.	2	34.	4	35.	5
36.	6	37.	4	38.	2	39.	2	40.	3
41.	(C)	42.	(D)	43.	(B)	44.	(B)	45.	(B)
46.	(B)	47.	(A)	48.	(B)	49.	(C)	50.	(B)